Foundations of Generative AI (ICT4403) — Assignment 3

Comparative English-to-French Machine Translation

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1 Problem statement and objective

Develop and compare four machine-translation systems for English-to-French parallel text: a basic encoder–decoder Transformer, mT5-small, a LLaMA-family decoder-only model, and Gemma-2-2B-it. All systems must translate the same held-out test set. Evaluation covers BLEU, chrF, training or fine-tuning time where applicable, inference time, parameter count, qualitative translation quality, sampling temperature, and responsible-AI risks.

2 Dataset description and source

- **Dataset:** OPUS Books, English–French parallel corpus.
- **Sources:** OPUS Books and Hugging Face dataset card.
- **Selection:** public book-domain parallel text; shuffled with seed 4403.
- **Subset:** 10,000 training pairs, 1,000 validation pairs, and 1,000 test pairs.
- **Cleaning:** whitespace normalization, removal of empty pairs, and filtering of source/target sentences longer than 250 characters.
- **Preprocessing:** word tokenization and learned vocabularies for the basic model; SentencePiece tokenization for mT5; model tokenizers and translation prompts for decoder-only models. The test sentences and references are shared across all four systems.

3 Models and experimental conditions

Model	Architecture	Adaptation and configuration
Basic Transformer	Encoder–decoder	PyTorch implementation from scratch; 3 encoder and 3 decoder layers, $d_{\text{model}} = 256$, 8 heads, feed-forward dimension 1024, dropout 0.1, AdamW, 8 epochs.
mT5-small	Encoder–decoder	`google/mt5-small`; prefix "translate English to French:"; supervised fine-tuning for 3 epochs.
TinyLlama 1.1B	Decoder-only	Open LLaMA-architecture variant; French instruction prompt and LoRA adaptation for 1 epoch.
Gemma-2-2B-it	Decoder-only	Instruction-tuned causal model; French instruction prompt and LoRA adaptation for 1 epoch.

Training is deterministic with seed 4403. The basic model uses teacher forcing and greedy decoding. Pretrained models use the same cleaned splits and capped sequence lengths. Evaluation uses SacreBLEU corpus BLEU and corpus chrF:

$$\text{BLEU} = \text{SacreBLEU corpus score}, \qquad \text{chrF} = \text{SacreBLEU character F-score}.$$

The decoder-only models are evaluated after reloading their LoRA adapters on the corresponding base checkpoints; generated prompt text is removed before scoring.

4 Quantitative results and execution status

The repository contains the complete reproducible pipeline. The numerical table below is intentionally not filled with estimated values: the final scores must be generated by executing the commands on a machine with the dependencies and model access.

Model	Arch.	BLEU	chrF	Train s	Infer ms/sent	Params
Basic Transformer	Enc–Dec	pending	pending	pending	pending	generated
mT5-small	Enc–Dec	pending	pending	pending	pending	∼300M
TinyLlama 1.1B	Decoder-only	pending	pending	pending	pending	∼1.1B
Gemma-2-2B-it	Decoder-only	pending	pending	pending	pending	∼2B

Run from the repository root:

```
python -m src.data
python -m src.basic_transformer
python -m src.train_pretrained mt5
```

```
python -m src.train_pretrained llama
python -m src.train_pretrained gemma
python -m src.evaluate --limit 1000
```

This writes `results/metrics.csv`, `results/metrics.json`, and the shared test translations to `results/samples/test_translations.csv`.

5 Model comparison

The basic Transformer is transparent and task-specific but must learn translation from the selected subset without pretraining. mT5 has a direct sequence-to-sequence inductive bias and multilingual pretraining, so it is expected to provide the strongest quality per parameter among the fine-tuned systems. Decoder-only models can use instruction following and LoRA, but their prompts increase input length and their larger parameter counts increase memory and inference cost. The final best-performing model must be selected from the generated BLEU and chrF results rather than from model size or expectation.

6 Sampling-temperature analysis

The default comparison uses deterministic generation for reproducibility. A separate analysis should generate identical probe sentences at $T \in \{0.2, 0.7, 1.0\}$:

$$P(w_i \mid w_{<i}) \propto \exp(z_i/T).$$

Low temperature produces stable, conservative translations but may repeat or omit content. Higher temperature increases lexical variation but can introduce grammatical errors, omissions, and fluent hallucinations. For production translation, deterministic decoding or a validated low temperature is preferable; temperature should not be used to hide low-confidence outputs.

7 Responsible and ethical AI

Translation systems may reproduce gender, regional, occupational, or dialect bias in training data. They can hallucinate fluent details, omit negation or numbers, and misrepresent idioms, honorifics, food, place names, and culturally specific references. Medical, legal, financial, identity, and personal text requires special caution because a plausible mistranslation can cause real harm.

Mitigations include representative and licensed data, language-pair and subgroup audits, named-entity and numeric-preservation checks, domain terminology constraints, human review for high-risk content, uncertainty escalation, privacy-preserving processing, and disclosure that outputs are machine-generated. BLEU and chrF measure overlap, not safety or cultural adequacy; qualitative review of saved samples is therefore required.

8 Conclusion

This work provides a reproducible comparison of encoder–decoder and decoder-only translation architectures on a shared English–French test set. The from-scratch Transformer establishes a transparent baseline, mT5 tests multilingual sequence-to-sequence fine-tuning, and TinyLlama and Gemma test parameter-efficient decoder-only adaptation. The final numerical conclusion should be inserted from `results/metrics.csv` after execution; no unrun or fabricated metric is reported here.

9 Submission details

- **Student:** Madhavan Komandur; Registration 230953298.
- **Developed system:** Comparative English-to-French Neural Machine Translation System.
- **Dataset:** OPUS Books — source link.
- **GitHub:** github.com/sltrtn/ICT4403-Assignment3-Machine-Translation.
- **Implementation:** `src/`, configuration in `configs/`, ethical analysis in `docs/`, and generated metrics/samples in `results/`.